

KARAN PRATAP SINGH RATHORE

✉ karanpratap1810@gmail.com | 📞 +91-7413087071 | 🌐 Portfolio | 🐙 GitHub | 🌐 LinkedIn

EDUCATION

Year	Degree/Certificate & Institute/Board	Institute	CGPA/Marks
2022–2026	B. Tech Indian Institute of Technology Jodhpur	IIT Jodhpur	8.58 / 10
2020–2021	Senior Secondary Examination RBSE	Vyas Public Senior Sec. School Bikaner	99.40%

WORK EXPERIENCE

- Reliance Industries Limited — Summer Intern** May 2025 – Jul 2025
- Developed a predictive maintenance pipeline using Python to analyze six months of industrial sensor data.
 - Engineered data processing workflows, feature extraction pipelines, and visualization tools for time-series analysis, validating predictions against engineering simulations.
- Sharp Career Technologies — Frontend Intern** Jun 2024 – Jul 2024
- Designed and implemented end-to-end product features for a full-stack hotel booking platform using React, JavaScript, REST APIs, and relational databases, collaborating on product requirements and feature delivery.
 - Built reusable React components, performed code refactoring, testing, debugging, and performance optimization while improving accessibility, usability, and maintainability.
- IICHe Student Chapter, IIT Jodhpur — Contributor** Aug 2023 – May 2024
- Designed, developed, and deployed the official chapter website using HTML/CSS/JS and React Framework, achieving a 50-point improvement in Google Lighthouse performance.
 - Led UI/UX implementation and feature rollouts for chapter events, delivering a responsive, mobile-friendly platform that increased student engagement.

PROJECTS

- GraphRAG – Movie Query System** | 2026
 - Built a production-grade GraphRAG knowledge retrieval system using Neo4j, Pinecone, Gemini, and AI tool-calling to answer complex natural language queries through hybrid graph and semantic search.
 - Designed a hybrid retrieval architecture combining graph traversal and vector similarity search, resolving entity disambiguation, semantic retrieval, and production-scale indexing for movies.
- CodeBulb – Copilot for VS Code** | Dec 2025 – Jan 2026
 - Built an AI-powered VS Code extension integrating Google Gemini and DeepSeek for real-time code generation, debugging, and project scaffolding.
 - Developed autonomous AI agents for file management, shell execution, and code generation using Node.js, publishing the extension on the VS Code Marketplace with a custom WebView interface.
- Fisher–Yates Algorithm Visualizer** |
 - Developed an interactive single-page application to visualize the Fisher–Yates shuffle algorithm using asynchronous JavaScript animations and dynamic DOM manipulation.
 - Implemented performance reporting with jsPDF and custom regex-based sanitization to generate downloadable execution reports.

COMPETITIONS / CONFERENCES

- Amazon ML Challenge 2025 — Smart Product Pricing** 2025
- Placed in the **top 1%** in the Smart Product Pricing Challenge (Amazon ML).
 - Engineered a multimodal pricing model combining text embeddings, CLIP-based image features, and metadata, improving SMAPE from a 61–66% baseline to 45–35%, a 20 percentage-point reduction.

TECHNICAL SKILLS

- Programming:** JavaScript (ES6+), TypeScript, Python, C++, SQL
- Frontend:** React.js, HTML5, CSS3, Bootstrap, Tailwind CSS
- Backend:** Node.js, Express.js, REST APIs, FastAPI, Backend System Design, API Design, Scalable Backend Development, Authentication
- AI Engineering:** Generative AI, LLM Applications, AI Agents, RAG, GraphRAG, Function Calling, Prompt Engineering, Semantic Search, Knowledge Graphs, Vector Databases, Embedding Models, LangChain, LangGraph, NumPy, Pandas
- Databases:** PostgreSQL, MySQL, Vector DB (Pinecone), Graph DB (neo4j)
- Developer Tools:** Git, GitHub, Linux, Docker, Postman
- Software Engineering:** Agile Development, Object-Oriented Design, API Design, Performance Optimization, Code Refactoring

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Probability & Statistics, Scientific Computations, Process Control & AI Applications.

POSITIONS OF RESPONSIBILITY

- Coordinator:** Career Guidance Cell, IIT Jodhpur May 2024 – Apr 2025
- Developer Head:** IICHe Student Chapter, IIT Jodhpur Sep 2024 – Dec 2024

AWARDS AND ACHIEVEMENTS

- Graduated with Department **Rank 2**, B.Tech in C.E., IIT Jodhpur 2026.